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In [1]: '''Program of univariate Linear Regression to fit a straight line using least square method
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import numpy as np
import matplotlib.pyplot as plt

X = np.array(eval(input()))
Y = np.array(eval(input()))

Xmean = np.mean(X)
Ymean = np.mean(Y)
num,den = 0,0
for i in range(len(X)):
    num += (X[i]-Xmean)*(Y[i]-Ymean)
    den += (X[i]-Xmean)**2
m = num/den
c = Ymean-m*Xmean

print (m, c)

Y_pred = m*X + c
print (Y_pred)

plt.scatter(X,Y)
plt.plot(X,Y_pred,color="red")
plt.show()

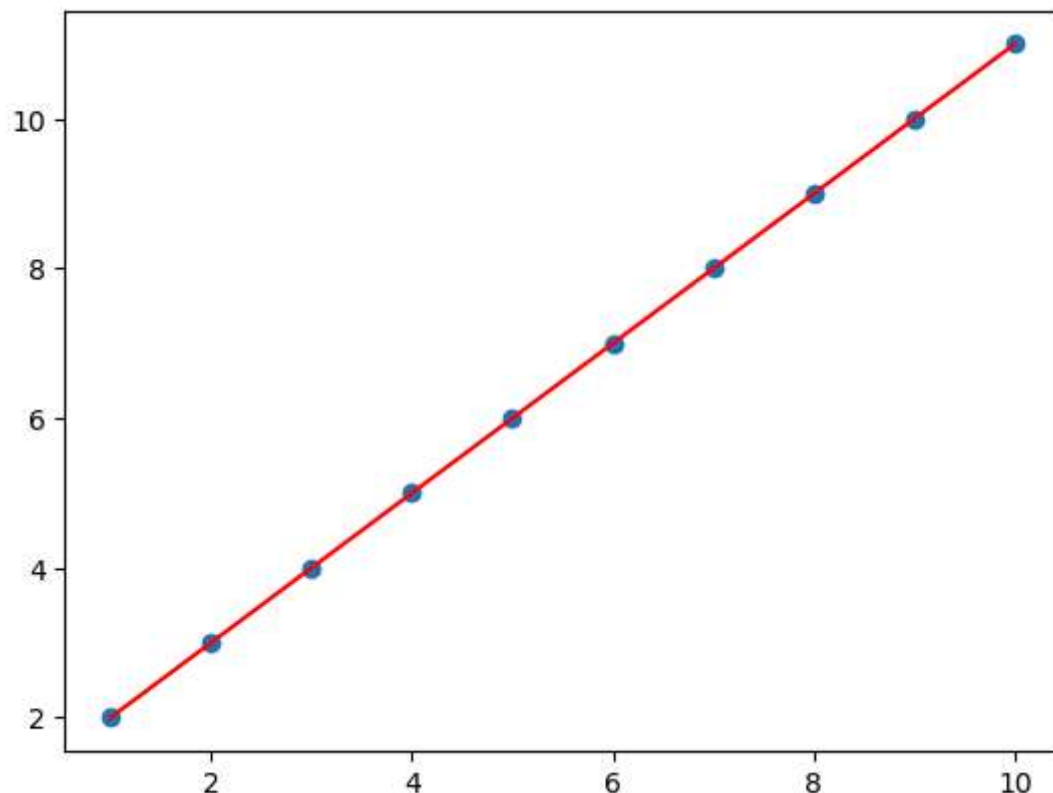
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1,2,3,4,5,6,7,8,9,10

2,3,4,5,6,7,8,9,10,11

1.0 1.0

[2. 3. 4. 5. 6. 7. 8. 9. 10. 11.]



In []: